

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

a. ① $H_0: \sigma_M^2 = \sigma_F^2$, $H_1: \sigma_M^2 \neq \sigma_F^2$

② $\alpha = 0.05$

③ Test statistic: $F = \frac{\sigma_M^2}{\sigma_F^2} \sim F_{N_M - K_M, N_F - K_F}$,

where the males have N_M observations and K_M parameters;

the females have N_F observations and K_F parameters;

④ $\hat{\sigma}_M^2 = \frac{SSE_M}{N_M - K_M} = \frac{97161.9174}{573} \approx 169.5670$

Realized statistic: $F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.5670}{12.024^2} \approx 1.1729$

⑤ Rejection Region: $\{F \mid F < F_{573, 419; 0.975} \text{ or } F > F_{573, 419; 0.025}\}$
 $= \{F \mid F < 0.8377 \text{ or } F > 1.1968\}$

⑥ Conclusion: Since $F^* = 1.1729$ doesn't fall in the rejection region, we can't reject the null hypothesis $\sigma_F^2 = \sigma_M^2$.

- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, \text{MARRIED} = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(e_i | \mathbf{x}_i, \text{MARRIED} = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add *FEMALE* to the wage equation as an explanatory variable, so that

$$\text{WAGE}_i = \beta_1 + \beta_2 \text{EDUC}_i + \beta_3 \text{EXPER}_i + \beta_4 \text{METRO}_i + \beta_5 \text{FEMALE} + e_i \quad (\text{XR8.6b})$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

b. ① $H_0: \sigma_M^2 = \sigma_S^2$, $H_1: \sigma_M^2 > \sigma_S^2$ (We write $\sigma_M^2 := \sigma_{\text{MARRIED}}^2$ and $\sigma_S^2 := \sigma_{\text{SINGLE}}^2$)

② $\alpha = 0.05$

③ Test statistic: $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} \sim F_{N_M - K_M, N_S - K_S}$,

where the married individuals have $N_M = 600$ observations and $K_M = 5$ parameters;
single individuals have $N_S = 400$ observations and $K_S = 5$ parameters;

④ $\hat{\sigma}_M^2 = \frac{\text{SSE}_M}{N_M - K_M} = \frac{100703.0471}{595} = 169.2488$

$\hat{\sigma}_S^2 = \frac{\text{SSE}_S}{N_S - K_S} = \frac{56231.0382}{395} = 142.3571$

Realized statistic: $F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} = \frac{169.2488}{142.3571} \approx 1.1889$

⑤ Rejection Region: $\{F \mid F > F_{595, 395; 0.05}\}$
 $= \{F \mid F > 1.1647\}$

⑥ Conclusion: Since $F^* = 1.1889$ fall in the rejection region,
we reject the null hypothesis $\sigma_M^2 = \sigma_S^2$,
so it is statistically significant that $\sigma_M^2 > \sigma_S^2$.

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

C. ① $W_{White} \sigma_i^2 = h(\alpha_1 + \alpha_2 EDUC_i + \alpha_3 EXPER_i + \alpha_4 METRO_i + \alpha_5 FEMALE_i)$

$H_0: \alpha_2 = \dots = \alpha_5$, H_1 : At least one of $\alpha_2, \dots, \alpha_5$ is different.

② $\alpha = 0.05$

③ Test statistic: $NR^2 \sim \chi^2_{S-1}$ ($S=5$)

④ Realized statistic: 59.03

⑤ Rejection region: $\{ \chi^2 \mid \chi^2 > \chi^2_{4,0.05} \} = \{ \chi^2 \mid \chi^2 > 9.488 \}$.

⑥ Since 59.03 falls in the rejection region, we reject the null hypothesis H_0 , and conclude that it is statistically significant that the error variance is related to at least one of the four variables. But it doesn't provide evidence about the issue in (b) since MARRIED is not in one of the four variables we discussed.

d. The variables are: EDUC, EXPER, METRO, FEMALE, $EDUC^2$, $EXPER^2$, (no $METRO^2$ and $FEMALE^2$ since they are both binary variables) and $C_2^4 = 6$ cross terms. So the degree of freedom is $4+2+6=12$

Test statistic: $NR^2 \sim \chi^2_{12}$

Realized statistic: 78.82

Critical value: $\chi^2_{12,0.05} = 21.026$

Since 78.82 is larger than 21.026, we conclude that it is statistically significant that the error variation is related to at least one of the cross-product terms of the four parameters.

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

e. The interval estimates of *EXPER*, *METRO*, *FEMALE* are narrower ; and the interval estimates of the intercept and *EDUC* are wider. It's not an inconsistency. Since it is not necessary that the interval estimates got wider or narrower when we using the robust standard error.

f. The *t*-value 1.0 shows that it is not statistically significant that $E[WAGE|MARRIED=1] \neq E[WAGE|MARRIED=0]$. This is not conflict with (b) because the result in (b) is concerning about the variance, not the conditional expectation.

- 8.16** A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- a. Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

```
Q. > library(POE5Rdata)
> data(vacation)
> ?vacation
>
> # Problem a
> mod1 <- lm(miles ~ income + age + kids, data=vacation)
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
    Min       1Q   Median       3Q      Max
-1198.14  -295.31   17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775  -2.306   0.0221 *
income       14.201      1.800    7.889 2.10e-13 ***
age          15.741      3.757    4.189 4.23e-05 ***
kids        -81.826     27.130   -3.016  0.0029 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

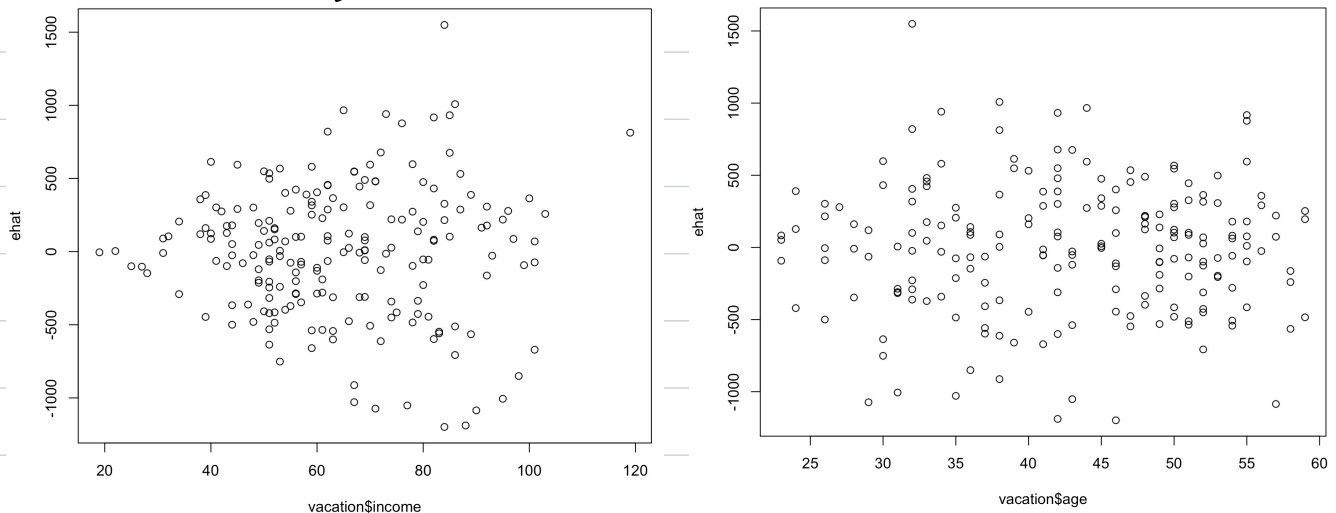
Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16

> b4 <- coef(smod1)[4,1]
> se4 <- coef(smod1)[4,2]
> se4
[1] 27.1296
> alpha <- 0.05
> df = df.residual(mod1)
> tcr <- qt(1-alpha/2, df)
> lowb4 <- b4 - tcr*se4
> upb4 <- b4 + tcr*se4
> cat("95% CI for beta4: [", lowb4, ", ", upb4, "]\n")
95% CI for beta4: [ -135.3298 , -28.32302 ]
```

⇒ 95% interval estimate for the effect of one more child on miles traveled is
[-135.3298 , -28.32302].

- b. Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?

```
> # Problem b
> ehat <- resid(mod1)
> plot(vacation$income, ehat)
> plot(vacation$age, ehat)
```



From the plot of the residuals versus income, we can observe that the heteroskedasticity is present

But from the plot of the residuals versus age, the heteroskedasticity is not obvious.

- c. Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

```
> # Problem c
>
> vacation_sorted <- vacation[order(vacation$income), ]
>
> mod_low <- lm(miles ~ income + age + kids, data = vacation_sorted[1:90, ])
> mod_high <- lm(miles ~ income + age + kids, data = vacation_sorted[111:200, ])
>
> SSE1 <- sum(resid(mod_low)^2)
> SSE2 <- sum(resid(mod_high)^2)
> df1 <- df.residual(mod_low)
> df2 <- df.residual(mod_high)
>
> GQ_stat <- (SSE2 / df2) / (SSE1 / df1)
> GQ_stat
[1] 3.104061
>
> p_value_GQ <- pf(GQ_stat, df2, df1, lower.tail = FALSE)
> p_value_GQ
[1] 1.64001e-07
```

$H_0: \sigma_{high}^2 = \sigma_{low}^2$, where σ_{high}^2 denotes the error variance of observations with high income (top 90), and σ_{low}^2 denotes the error variance of observations with low income (low 90).

$H_1: \sigma_{high}^2 \neq \sigma_{low}^2$

- d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

```
> # Problem d
> # install.packages("car")
> # install.packages("lmtest")
> library(car)
> library(lmtest)
>
> cov1 <- hccm(mod1, type = "hc1")
> robust_result <- coeftest(mod1, vcov.= cov1)
> robust_result

t test of coefficients:

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.5480    142.6548  -2.7447 0.0066190 **
income       14.2013      1.9389   7.3246 6.083e-12 ***
age          15.7409      3.9657   3.9692 0.0001011 ***
kids        -81.8264     29.1544  -2.8067 0.0055112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> b4_robust <- robust_result["kids", "Estimate"]
> se4_robust <- robust_result["kids", "Std. Error"]
>
> tcr <- qt(1-alpha/2, df.residual(mod1))
>
> low_d <- b4_robust - tcr * se4_robust
> up_d <- b4_robust + tcr * se4_robust
> cat("Robust 95% CI for beta4: [", low_d, ", ", up_d, "]\n")
Robust 95% CI for beta4: [ -139.323 , -24.32986 ]
```

The 95% interval estimate using heteroskedasticity is $[-139.323, -24.32986]$. The length of this interval is wider than the interval in (a).

- e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

```
> # Problem e
>
> n <- nrow(vacation)
>
> vacation$miles_w <- vacation$miles / vacation$income # y*
> vacation$inv_income <- 1 / vacation$income # x1*
> vacation$age_w <- vacation$age / vacation$income # x3*
> vacation$kids_w <- vacation$kids / vacation$income # x4*
>
> mod_gls <- lm(miles_w ~ inv_income + age_w + kids_w, data = vacation)
> summary(mod_gls)

Call:
lm(formula = miles_w ~ inv_income + age_w + kids_w, data = vacation)

Residuals:
    Min       1Q   Median       3Q      Max
-15.1907  -4.9555   0.2488   4.3832  18.5462

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   13.947      1.481    9.420 < 2e-16 ***
inv_income  -424.996    121.444  -3.500 0.000577 ***
age_w         16.717      3.025   5.527 1.03e-07 ***
kids_w        -76.806     21.848  -3.515 0.000545 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.765 on 196 degrees of freedom
Multiple R-squared:  0.1469, Adjusted R-squared:  0.1338
F-statistic: 11.25 on 3 and 196 DF, p-value: 7.644e-07

> # --- conventional GLS ---
> gls_coef <- coef(summary(mod_gls))
> b4_gls <- gls_coef["kids_w", "Estimate"]
> se4_gls <- gls_coef["kids_w", "Std. Error"]
>
> tcr_gls <- qt(1-alpha/2, df.residual(mod_gls))
>
> low_e_conv <- b4_gls - tcr_gls * se4_gls
> up_e_conv <- b4_gls + tcr_gls * se4_gls
> cat("GLS conventional 95% CI for beta4: [", low_e_conv, ", ", up_e_conv, "]\n")
GLS conventional 95% CI for beta4: [ -119.8945 , -33.71808 ]
>
> # --- Robust GLS ---
> cov_gls <- hccm(mod_gls, type = "hcl1")
> robust_gls <- coeftest(mod_gls, vcov = cov_gls)
> b4_gls_r <- robust_gls["kids_w", "Estimate"]
> se4_gls_r <- robust_gls["kids_w", "Std. Error"]
>
> low_e_rob <- b4_gls_r - tcr_gls * se4_gls_r
> up_e_rob <- b4_gls_r + tcr_gls * se4_gls_r
> cat("GLS robust 95% CI for beta4: [", low_e_rob, ", ", up_e_rob, "]\n")
GLS robust 95% CI for beta4: [ -121.4134 , -32.19919 ]
>
> cat("(a) OLS conventional: [", lowb4, ", ", upb4, "]\n")
(a) OLS conventional: [ -135.3298 , -28.32302 ]
> cat("(d) OLS robust: [", low_d, ", ", up_d, "]\n")
(d) OLS robust: [ -139.323 , -24.32986 ]
> cat("(e) GLS conventional: [", low_e_conv, ", ", up_e_conv, "]\n")
(e) GLS conventional: [ -119.8945 , -33.71808 ]
> cat("(e) GLS robust: [", low_e_rob, ", ", up_e_rob, "]\n")
(e) GLS robust: [ -121.4134 , -32.19919 ]
```

Conventional GLS: [-119.8945, -33.71808]

Robust GLS: [-121.4134, -32.19919]

These two intervals are both wider than the intervals in (a) and (d).